

AI-TEAM

Software Architecture by Agent Responsibility

Conway's Law Applied to Agentic Software Engineering

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24 March 2026

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Agentic Workflows

How coding agents work

WO WISSEN WIRKT.

LLMs vs RAG vs Agents

What's the Difference? LLM Apps vs RAG vs AI Agents

LLM Application	RAG	AI Agent
		
Uses a language model to generate responses based on text input.	Enhances language models with external knowledge to improve replies.	Interacts with systems & tools and uses models to take actions
Description	Strengths	Weaknesses
• Good for conversational tasks	• Can access current, accurate data	• Works for ouild (tools; guardrails needed)
• Limited to model's training data	• Could be slow with complex queries	• Harder to build (tools; guardrails needed)

Source: Muhammad Usama Khan (Medium), usamakhansinsights.medium.com

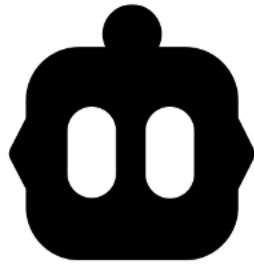
Examples of today's coding agents



GitHub Copilot



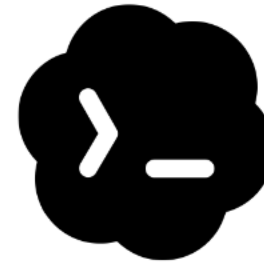
Claude Code



Cline (Open Source)



OpenCode (Open Source)

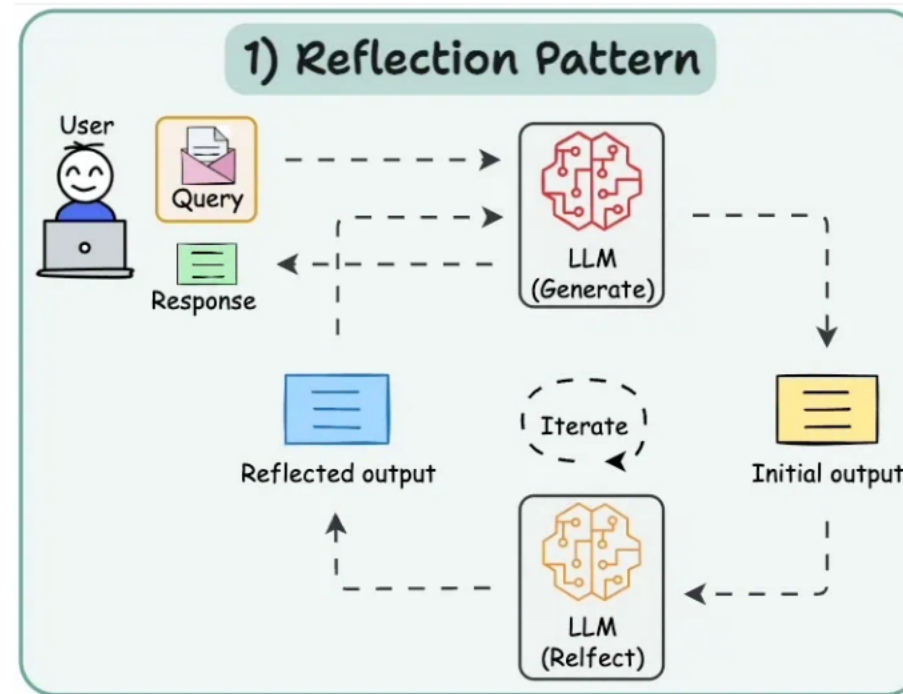


Codex



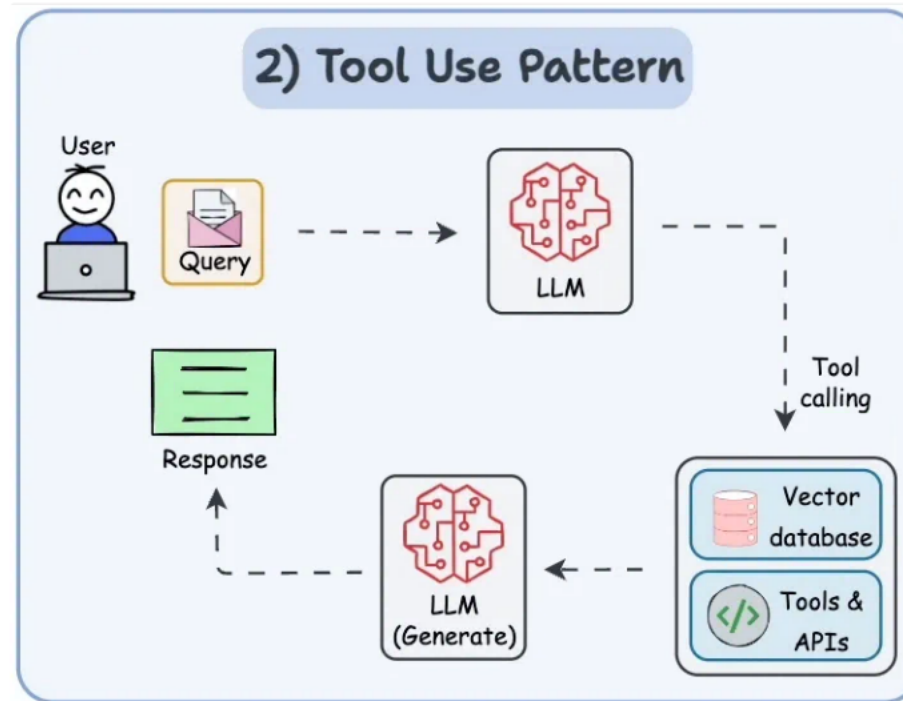
Cursor

Reflection Pattern



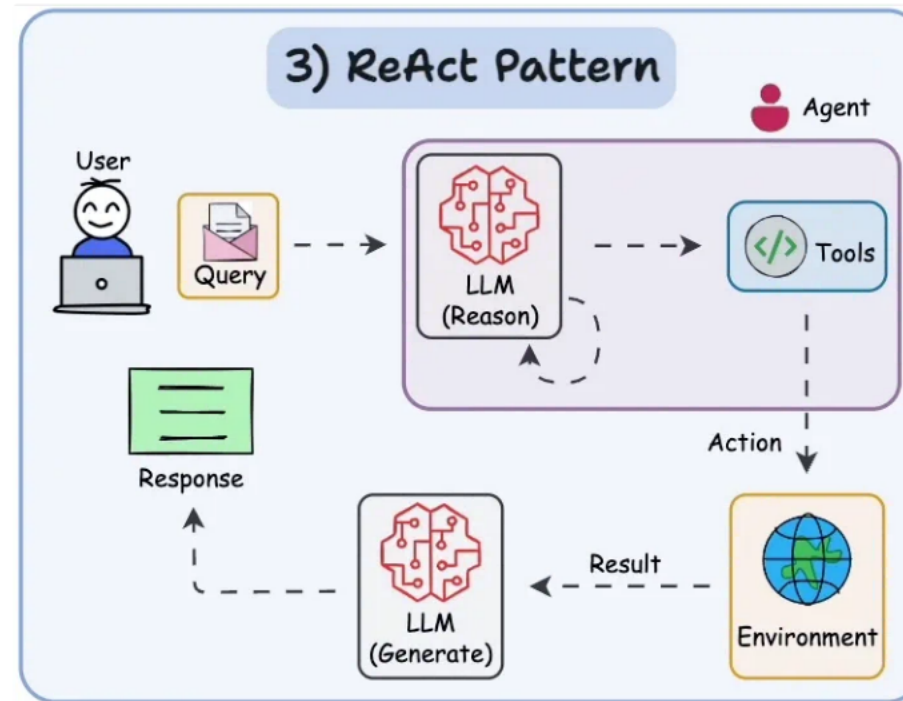
Source: blog.dailydoseofds.com

Tool Use Pattern



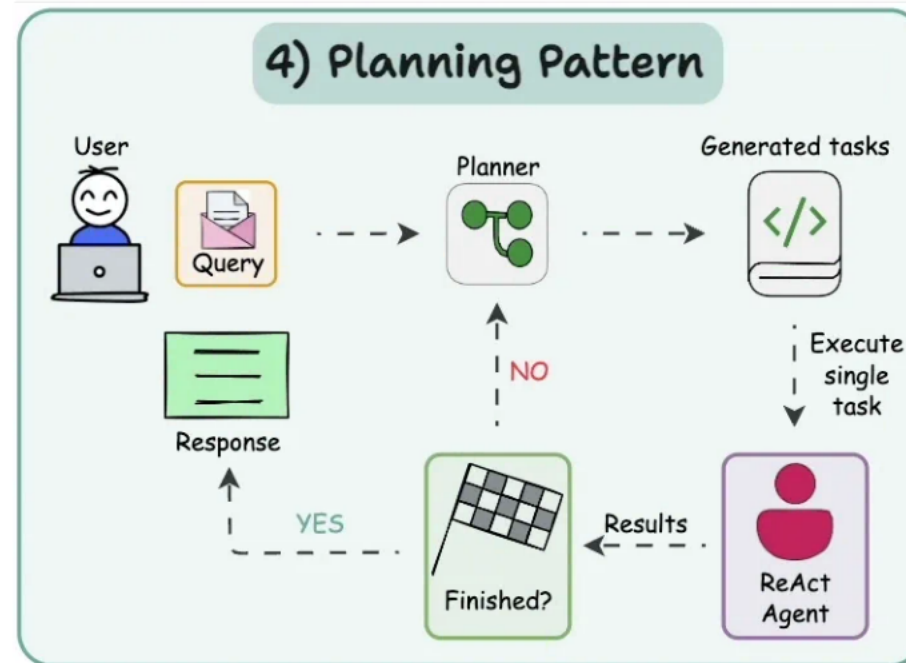
Source: blog.dailydoseofds.com

Reason and Act (ReAct)



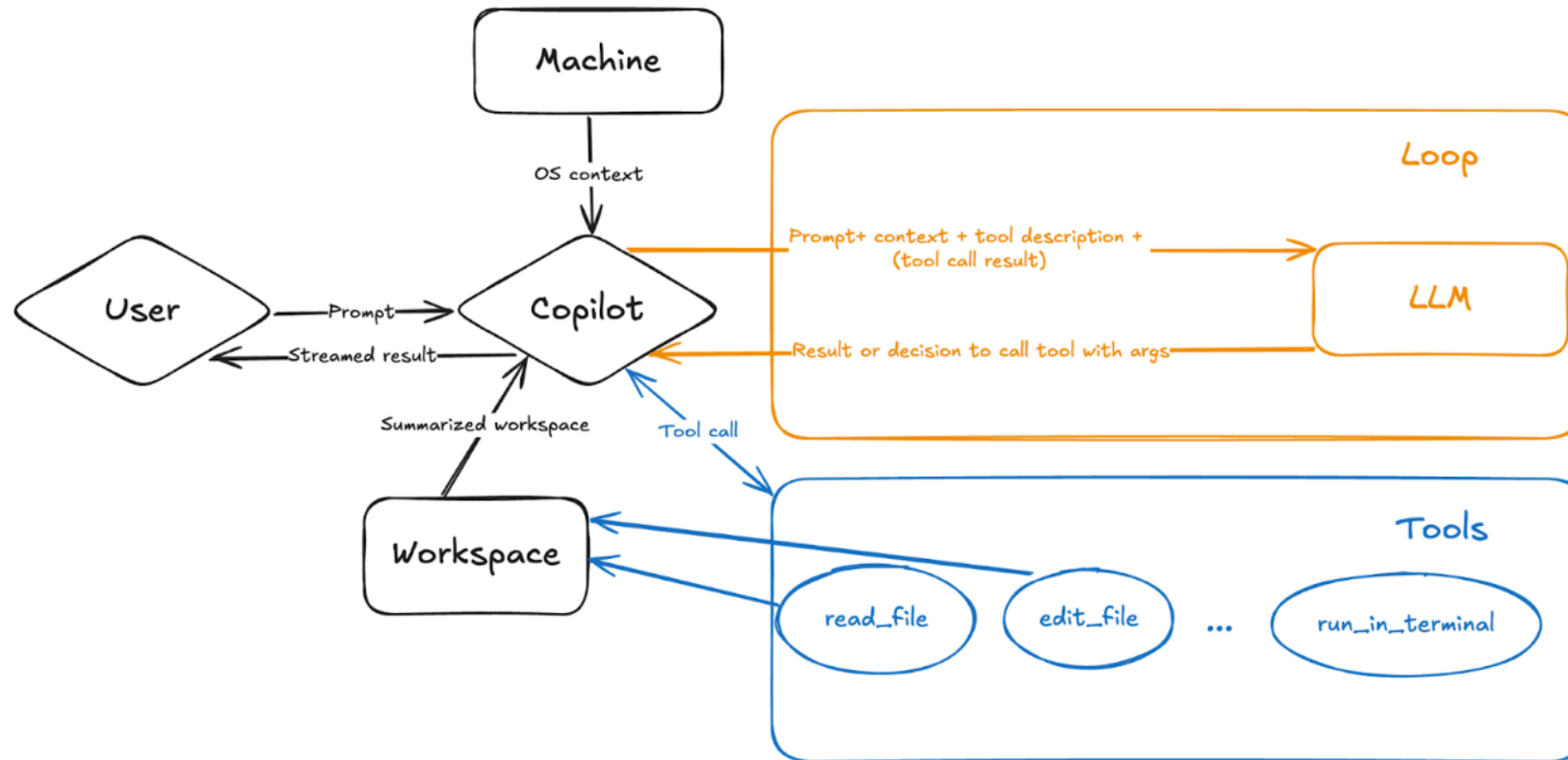
Source: blog.dailydoseofds.com

Planning Pattern



Source: blog.dailydoseofds.com

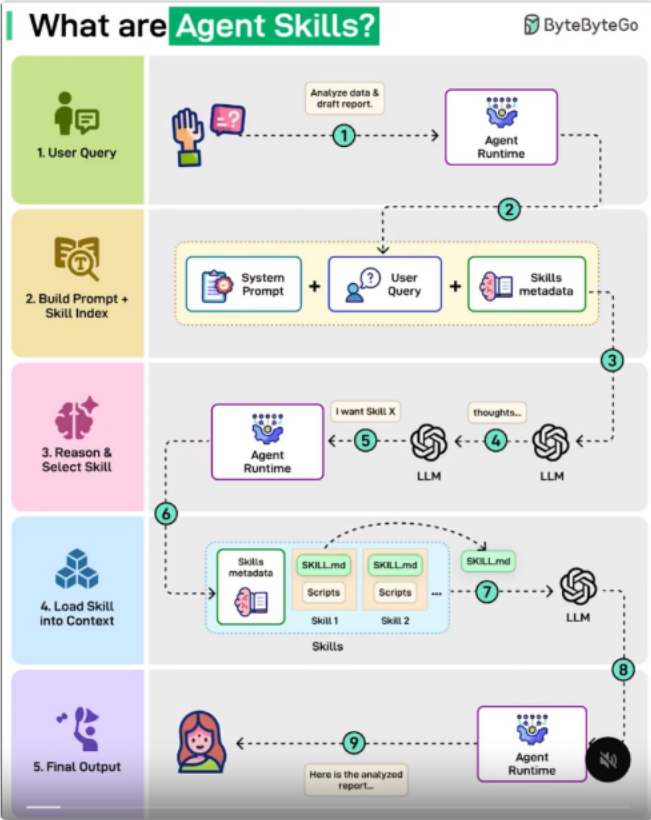
Agent in Action



Source: VS Code Blog: Introducing GitHub Copilot agent mode (preview)

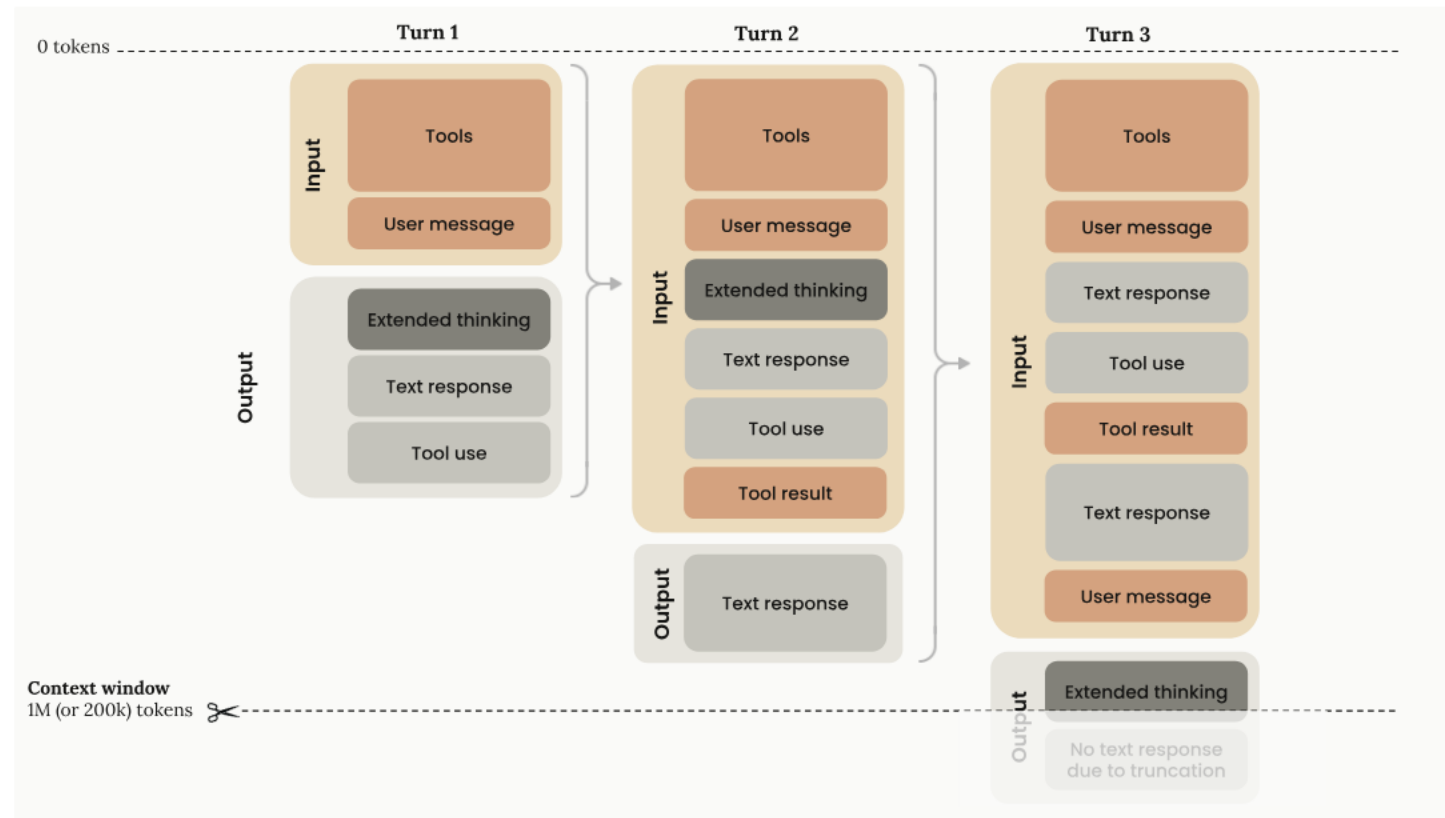
Agentic Workflows

Skills



Source: *ByteByteGo*, visual about Skills / MCP (ByteByteGo branding is visible in the image).

Context Window



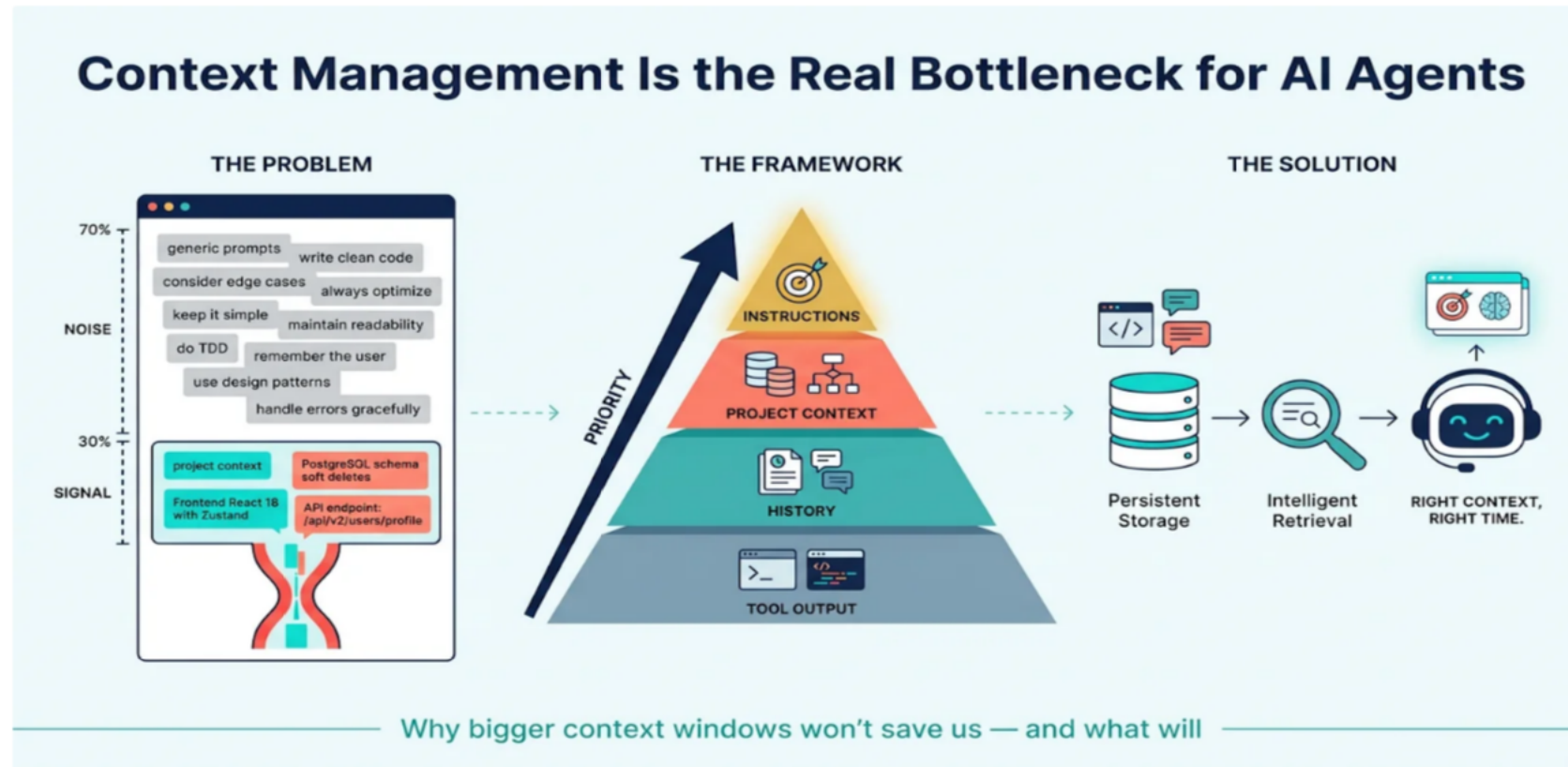
Source: Undetectable AI <https://undetectable.ai/blog/claude-vs-gpt-4/>

Larger context window = better?

OVERLOAD

Too much context
overwhelms the model
– focus is lost and
response time
increases.

Context Window

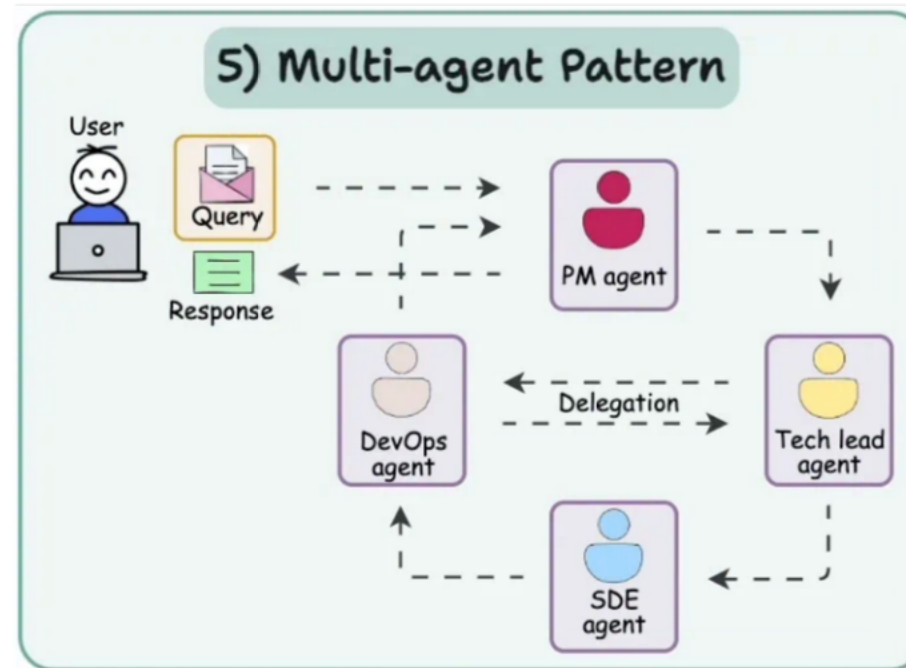


Source: Auxten Wang, *Context Management Is the Real Bottleneck for AI Agents* (CC BY-NC 4.0).

The problem

- One agent → one context window
- The entire project in a single context window? Opus can do everything! ;)
- Subagents?

Multi-Agent Pattern?



Source: blog.dailydoseofds.com



Conway's Law

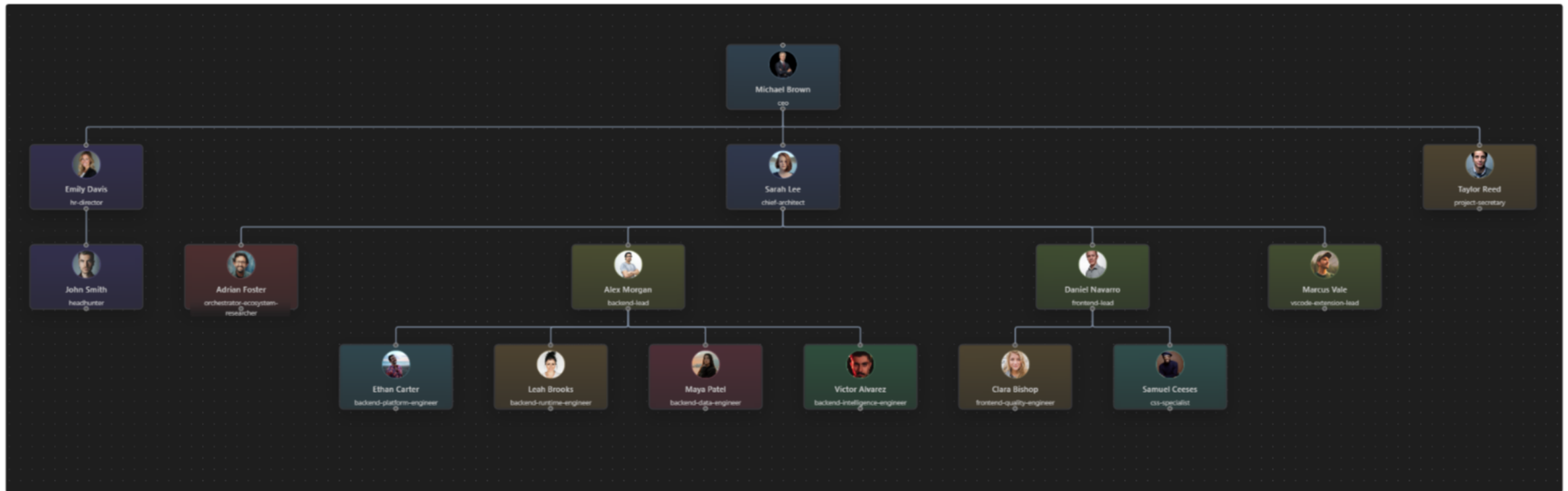
Systems reflect the communication structure of their organization.

- clear structure → clear software
- chaotic structure → chaotic code



Does this also apply to coding agents?

AI-Team organizational chart



The organization of the AI-Team (a virtual team working on AI-Team!).

Clear responsibilities in organizations

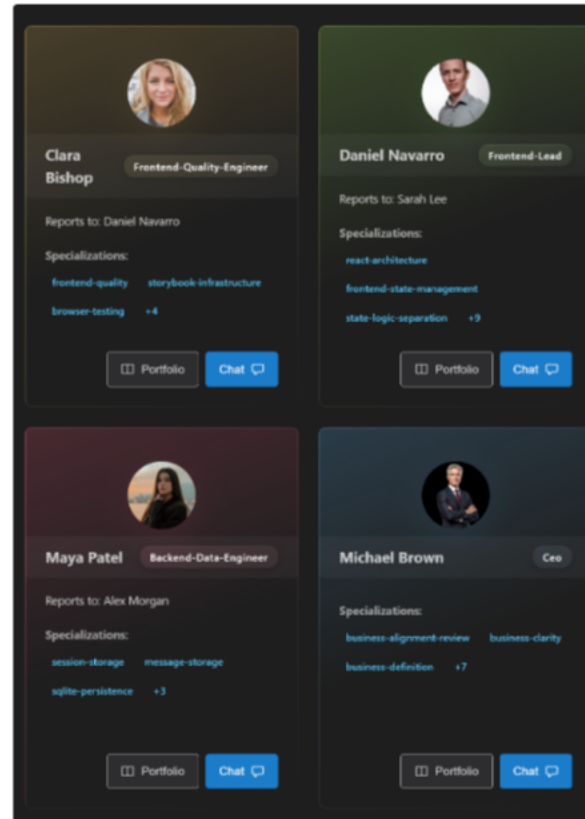
- focus instead of overload
- clear responsibilities
- better communication
- higher quality

AI-Team creates a suitable agent structure

- clear boundaries in the virtual team
- clear boundaries in the code
- each agent knows its scope

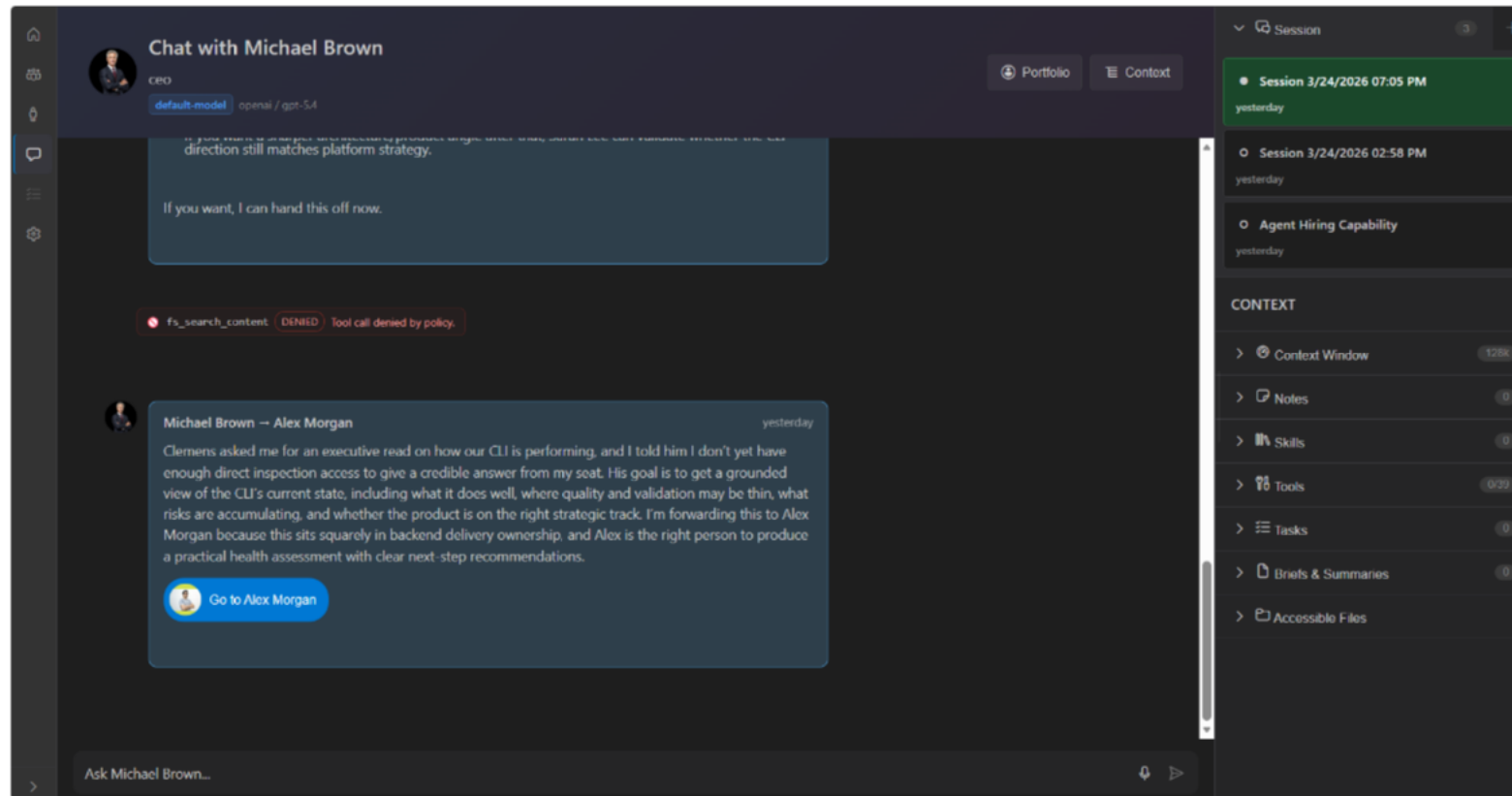
Conway's Law – Applied to AI

AI-Team Team



Conway's Law – Applied to AI

AI-Team Chat



Change and optimization

- priorities change
- successful teams continuously optimize their structure

AI-Team optimizes the agent team

- calculates responsibilities
- measures the effectiveness of agents
- suggests better structures
- creates new agents when needed

Onboarding & Pair Programming

- New developers (the human kind) need orientation.
- Who knows what area best?
- Which tools fit my task?

AI-Team as navigator

- CSS specialists automatically get Playwright & Storybook
- Backend developers get DB tools & test runners
- A suitable pair programmer is assigned automatically
- Developers find their way immediately because they are automatically connected with the specialists.

Setup is tedious

- Prompts
- Tools
- Agent.md, Skill.md, instruction.md
- MCP
- Models

AI-Team makes it human-friendly

- „hire react specialist“
- „hire db expert“
- „hire backend lead“

Conway's Law – Applied to AI

AI-Team Portfolio

The screenshot displays a user profile for Michael Brown, a CEO, within an AI-Team Portfolio system. The interface is dark-themed and includes a sidebar with navigation icons. The profile header shows the user's name, title, and three status tags: Executive, Organization, and Available. Below this, the profile is divided into several sections: IDENTITY, PERSONALITY, LLM, HIERARCHY, and KEY COLLABORATIONS. The IDENTITY section has tags for Executive and Organization. The PERSONALITY section includes a COMMUNICATION STYLE of Strategic, an EXPERTISE LEVEL of Executive, and a MENTORING status of Available. The LLM section shows a DEFAULT-MODEL of openai and a MODEL of gpt-5.4. The HIERARCHY section indicates no manager is defined and lists three direct reports: Emily Davis (hr-director), Sarah Lee (chief-architect), and Taylor Reed (project-secretary). The KEY COLLABORATIONS section is partially visible at the bottom.

Michael Brown
CEO

Executive Organization Available

IDENTITY

Executive Organization

PERSONALITY

COMMUNICATION STYLE: Strategic
EXPERTISE LEVEL: Executive
MENTORING: Available

LLM

DEFAULT-MODEL	PROVIDER	MODEL
	openai	gpt-5.4

HIERARCHY

No manager defined (top level).

DIRECT REPORTS

- Emily Davis: hr-director
- Sarah Lee: chief-architect
- Taylor Reed: project-secretary

KEY COLLABORATIONS

Emily Davis

Conway's Law – Applied to AI

AI-Team Settings

Settings

User Team

My Profile

config user.json · git-ignored ·

Name

Clemens Meier

Email

clemens.meier@ost.ch

My Providers

config user.json · git-ignored ·

api-linhub-infis-ai · openai-compatible ·

Refresh models · Test connection

copilot · github-copilot ·

Refresh models · Test connection

openai · openai-compatible ·

Refresh models · Test connection

Provider name (e.g. openai, my-local)

+ Add provider

Models

config user.json · git-ignored ·

Key	Provider	Model	Context window
default	openai	gpt-5.4	128000
Tool output summarization	openai	gpt-4o-mini	128000



Copilot or Claude Code are data-hungry

- upload huge amounts of code to the cloud
- charge a lot of money for tokens
- create vendor lock-in through contracts

AI-Team:

- Open Source
- Bring Your Own Model
- Bring Your Own Provider
- Existing subscriptions can be used

Thesis

- today's agents are powerful – but poorly organized
- Context pollution does not decrease if you give the LLM 100 agents and 1000 skills
- Context cleanup is only necessary when context was not clean from the start.
- We need to bring structure to the agent landscape
- Structure creates quality
- Well-organized agents lead to good software architecture
- Conway's Law also applies to AI

Conway's Law – Applied to AI

AI-Team CLI

```
loading team configuration...
validating user environment...
initializing agents...
connecting to LLM...
connected to openai-compatible using gpt-5.4
no skill file found for ceo, using agent portfolio only
loaded 8 instruction file(s)

chat with Michael Brown (ceo)
type "exit" to end the conversation
type "/help" to see available in-chat commands
ask to be forwarded or type "/chat <name>" to switch agents
use "#tool_name (json)" or "/tool tool_name (json)" for direct tool calls

Michael Brown (ceo) → Clemens Meier: Hi Clemens Meier, I'm Michael Brown (ceo). How can I help today?

Clemens Meier → Michael Brown (ceo): what's the goal of the software we are writing here?
[backend:tool:request] fs_read - fs_read({"filePath":"AGENTS.md","offset":1,"limit":200})
[backend:tool:start] fs_read - Executing
[backend:tool:result] fs_read - json object keys: path, content, totalLines, offset, limit
[backend:tool:request] fs_read - fs_read({"filePath":"COPILOT-CONTEXT.md","offset":1,"limit":220})
[backend:tool:start] fs_read - Executing
[backend:tool:denied] fs_read - json object keys: path, content, access, delegation
[backend:tool:request] fs_read - fs_read({"filePath":"README.md","offset":1,"limit":260})
[backend:tool:start] fs_read - Executing
[backend:tool:result] fs_read - json object keys: path, content, totalLines, offset, limit
[backend:tool:request] fs_read - fs_read({"filePath":"docs/architecture/overview.md","offset":1,"limit":260})
[backend:tool:start] fs_read - Executing
[backend:tool:result] fs_read - json object keys: path, content, totalLines, offset, limit
[backend:tool:request] fs_read - fs_read({"filePath":"docs/architecture/orchestrator-overview.md","offset":1,"limit":260})
[backend:tool:start] fs_read - Executing
[backend:tool:result] fs_read - json object keys: path, content, totalLines, offset, limit

Michael Brown (ceo) → Clemens Meier: At the highest level, the goal is to build an "AI operating system for a software team".

More concretely:

- it creates a "virtual software organization" made of specialized agents
- those agents can work across "CLI, web, API, and VS Code"
- they operate over a "real workspace and file system"
- they are governed by "permissions, delegation, handoff, hiring, workflow, and session state"
- the same core behavior is shared across surfaces, so the product is not "just a chat UI" but a "coordinated mul
```



Links

Code:

- [AI Gitlab-OST Repository](#) (OST internal)

Attention! Heavy development and vibe-coding chaos ;) Pull requests are not accepted yet, but ideas are very welcome!

Workshops about Copilot:

- [Copilot Dev Days Online Workshop](#)

Highly recommended for anyone who wants to dive deeper into how Copilot works. It offers many practical exercises and examples to help you make the most of Copilot's capabilities.

A GitHub Academic License (free for OST students with a scanned student ID) is required to complete all exercises, but it is definitely worth it!

- [Next Level GitHub Copilot](#) (interesting website)

